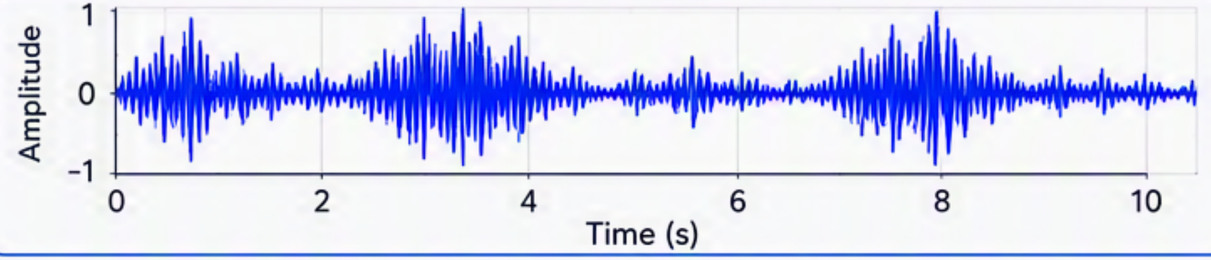
1. Time Signal





- Raw acoustic / vibration time-series signal
- Amplitude vs time

2. FFT

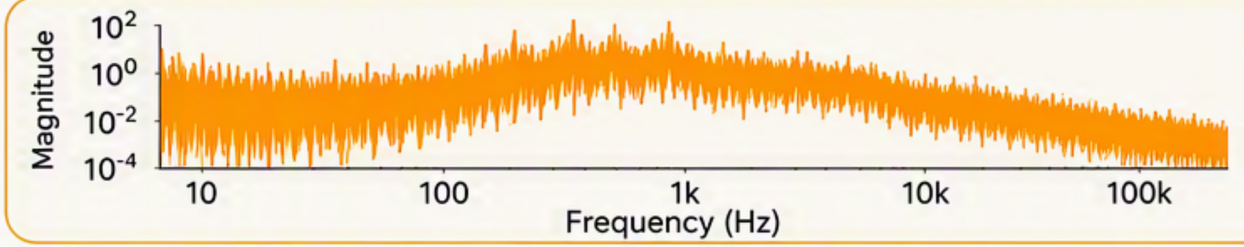




- Convert time signal to frequency domain
- FFT applied to each time window

3. Magnitude Spectrum

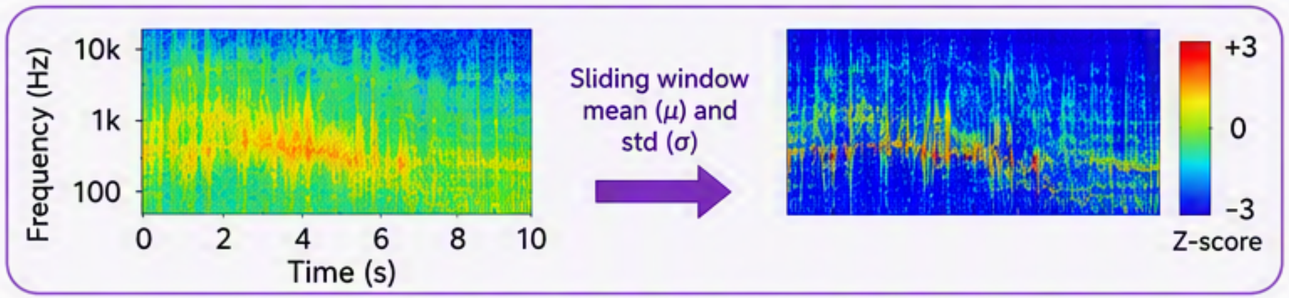




- Magnitude of FFT
- Amplitude vs frequency (log scales often used)

4. Sliding-Window Z-Score Normalisation

$$z = \frac{x - \mu}{\sigma}$$



- Compute mean and std-dev in a sliding window for each frequency bin
- Apply z-score normalisation

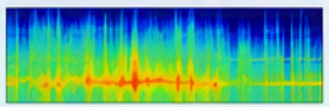
5. Colour Mapping

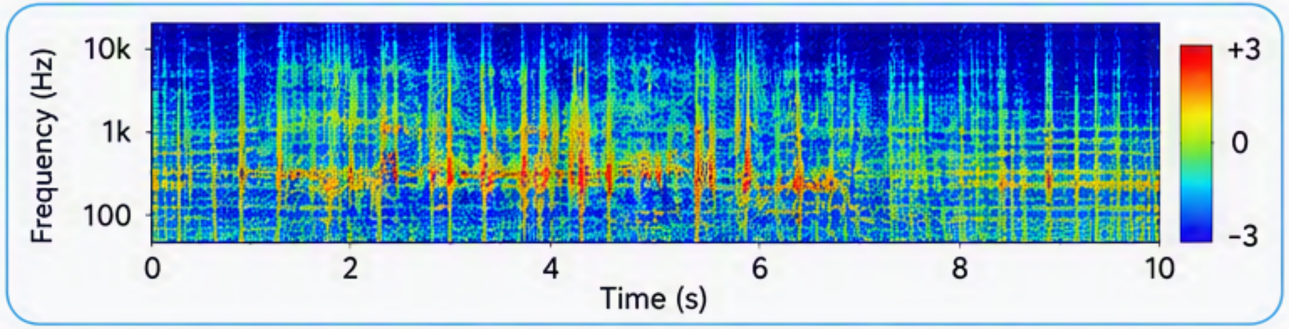




- Map z-score values to colours
- Enhances contrast and highlights features

6. Displayed LOFARgram / Spectrogram





- Final LOFARgram / Spectrogram
- Frequency vs time
- Colour shows normalised intensity (z-score)